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Role of Thiamine in Alzheimer's Disease

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Abstract

Alzheimer's disease (AD) is the most common form of dementia in elderly individuals and is associated with progressive neurodegeneration of the human neocortex. Thiamine levels and the activity of thiamine-dependent enzymes are reduced in the brains and peripheral tissues of patients with AD. Genetic studies have provided the opportunity to determine what proteins link thiamine to AD pathology (ie, transketolase, apolipoprotein E, α-1-antitrypsin, pyruvate dehydrogenase complex, *p53*, glycogen synthetase kinase-3β, *c-Fos* gene, the *Sp1* promoter gene, and the poly(ADP-ribosyl) polymerase-1 gene). We reviewed the association between histopathogenesis and neurotransmitters to understand the relationship between thiamine and AD pathology. Oral thiamine trials have been shown to improve the cognitive function of patients with AD; however, absorption of thiamine is poor in elderly individuals. In the early stage of thiamine-deficient encephalopathy (Wernicke's encephalopathy), however, parental thiamine has been used successfully. Therefore, further studies are needed to determine the benefits of using parental thiamine as a treatment for AD.

Keywords

thiamine, Alzheimer's disease, dementia, vitamin B_1

Introduction

Glucose is required for energy generation in the brain. Nearly 30% of brain glucose undergoes complete oxidation through the mitochondrial tricarboxylic acid cycle.[1] Thiamine is transported to the brain and is phosphorylated to thiamine diphosphate (TDP) by thiamine pyrophosphorylase. In the neuronal metabolism of glucose, TDP is an essential coenzyme for mitochondrial pyruvate, α-ketoglutarate dehydrogenases (KGD) complexes, and cytosolic transketolase (TK).[2,3] In thiamine deficiency, the levels of thiamine-dependent and non-thiamine-dependent enzymes (succinate and malate dehydrogenase) in the tricarboxylic acid cycle are reduced in the mouse brain.[4] Similarly, the activity of brain KGD is also reported to decrease in Alzheimer's disease (AD),[5] suggesting that thiamine may have a role in the pathogenesis of AD. Therefore, we review the role of thiamine in AD pathology.

The Relationship Between Thiamine and AD

Nutritional Factors

Disturbance of glucose metabolism is a prominent characteristic in the brains of patients with AD, and type 2 diabetes mellitus has been identified as a risk factor for AD.[6-8] Low blood thiamine levels, erythrocyte TK activity, and high erythrocyte thiamine pyrophosphate (TPP) activity have been documented in diabetic patients.[9,10] In addition, plasma thiamine level has been shown to be decreased by 76% in patients with type 1 and 75% in patients with type 2 diabetes and associated with increased renal clearance and fractional excretion of thiamine.[11] Similarly, low plasma thiamine levels have also been reported in patients with AD.[12-14] In AD brains, the activities of the KGD were reduced by more than 75% and those of TK by more than 45%; significant abnormalities of TK were identified in red blood cells and cultured fibroblasts of patients with AD.[15] These findings suggested that thiamine may have a role in the pathogenesis of AD.

Genetic Factors

Genetic studies provide an excellent opportunity to link molecular variations with epidemiological data. DNA sequences variations such as polymorphisms have modest and subtle biological effects. Receptors play a crucial role in the regulation of cellular function, and small changes in their structure can influence intracellular signal transduction pathways.

Transketolase has been used to assess thiamine activity in mammalian tissue. Structural abnormalities of TK occur in a high

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proportion of patients with AD.[16,17] Fibroblasts from patients with AD propagated at different pH values, within a range of 7.3 to 7.8, have been shown to exhibit TK abnormalities.[18] In addition, proteolytic cleavage has been shown to decrease TK activity[19] and might be used as biochemical marker in fibroblasts of patients with AD. Similarly, TK variants have been found in patients with thiamine-deficient encephalopathy.[20] The TK extracted from the fibroblasts of thiamine-deficient alcoholic patients and their sons has been shown to have a decreased affinity for TPP.[21,22] Finally, Heap et al[23] investigated the affinity of erythrocyte TK for its cofactor in alcoholics with cognitive impairment. They reported that, despite adequate nutrition, this patient subgroup has altered TK-binding. These reports, however, did not rule out the alcoholic effect on the TK.

Apolipoprotein E (*ApoE*) is a major genetic factor identified in AD. Carriers of at least 1 *ApoE* ε4 allele have an increased risk of developing AD.[24,25] *ApoE* has been shown to be significantly altered in the cerebrospinal fluid (CSF) of patients with AD.[26] In addition, capillary cerebral amyloid angiopathy has been identified as a distinct *ApoE* ε4-associated subtype of sporadic AD,[27] which may determine the clinical phenotype of AD.[28] Patients expressing this *ApoE* genotype are known to have significant impairment in memory retention. Conversely, *ApoE* was not found to be a risk or a protective factor for AD in an Ecuadorian population.[29] On the other hand, Muramatsu et al[30] reported that the frequency of the *ApoE* ε4 allele was significantly higher in patients with thiamine-deficient dementia.

α-1-Antitrypsin (*ATT*) has been identified as a potential biomarker in the plasma of patients with AD.[31] Higher levels of *ATT* in CSF were associated with the clinical diagnosis of AD.[26,32] In a study consisting of 1136 patients presenting with cognitive disorders, polymorphisms in *ApoE*, the gene for hemochromatosis and *ATT* were present in up to 40% of the patients.[33] Similarly, *ATT* polymorphism is associated with increased incidence of thiamine deficiency.[34]

Genetic defects in pyruvate dehydrogenase complex (PDHC) are known to cause lactic acidosis, neurological deficits, and premature death.[35] Patients with these defects show reduced activity of PDHC and pyruvate dehydrogenase (PDH) E1-alpha subunit and decreased affinity of PDHC for TPP.[36,37] Thiamine treatment is very effective in some patients with PDHC-deficiency.[37,38] Thiamine regulates the expression of enzymes that require thiamine as a cofactor and thiamine deficiency has been shown to reduce the messenger RNA (mRNA) levels of TK and PDH.[39]

There are numerous potential gene products that are transcriptionally activated by *p53* and are involved in cell cycle arrest or apoptosis.[40] In cellular models of AD, *p53* has been found to be undergoing conformational changes that make cells less vulnerable to stressors or genotoxic insults.[41,42] DNA damage has been shown to increase *p53* DNA-binding activity. Furthermore, TDP has been shown to inhibit *p53* binding, and thiamine has been shown to inhibit intracellular *p53* activity.[43] Recently, *p53* and glycogen synthetase kinase-3β (GSK3β) have been linked to AD.[44,45] The GSK3β is a protein kinase involved in many physiological processes, for example, cell structure, metabolism, gene expression, and apoptosis. In addition, the interaction of *p53* and GSK3β has been shown to lead to an increased activity of both proteins. Exposure to pyrithiamine, an anti-thiamine compound, has also been shown to increase β-amyloid protein accumulation and GSK3 activity in the brain.[46] In an animal AD model, benfotiamine was found to improve the cognitive function, reduce amyloid deposit, and suppress GSK3 activity.[47] These findings suggested a role of thiamine in controlling the activity of *p*53 in patients with AD.

The proto-oncogene protein *c-Fos* codes for a nuclear protein involved in growth-related transcriptional control. Activation of c-*Fos* has been suggested to contribute to β-amyloid protein-induced neurotoxicity in AD,[48] and over-expression of *c-Fos* mRNA has been reported in AD.[49,50] Similarly, *c-Fos* expression has been shown to increase with the progression of thiamine deficiency and has been associated with neuronal loss in the thalamus and inferior colliculus of the thiamine-deficient rat.[51-53]

Thiamine uptake in the human intestine occurs via a specialized carrier-mediated mechanism. The human thiamine transporters were found to be expressed in the human intestine, which are regulated via *Sp1* promoter elements.[54] *Sp1* transcription factor may be involved in regulating the expression of several AD-related proteins. Abnormal *Sp1* transcription factor has been reported in AD.[55,56] The regulatory region of the amyloid precursor protein (APP) gene contains sites recognized by the Sp1 transcription factor, which has been shown to be required for the regulation of APP and Aβ.[57] $BACE_1$, the major β-secretase involved in cleaving APP promoter contains a functional Sp1 response element, and overexpression of the Sp1 transcription factor potentiates *BACE* gene expression and APP processing to generate Aβ.[58] Sp1 and signaling mother against decapentaplegic peptide (Smad) transcription factors cooperate to potentiate transforming growth factor-β-dependent activation of APP.[59] These findings suggested a role of Sp1, the thiamine homeostasis, in AD.

Poly(ADP-ribosyl) polymerase (PARP) is a nuclear protein that contributes to both neuronal death and survival under stress condition. Poly(ADP-ribosyl) polymerase cleavage is enhanced in peripheral blood mononuclear cells from patients with mild cognitive impairment.[60] Enhanced-PARP activity is reported in AD and is suggested as a marker for AD.[61,62] Poly(ADP-ribosyl) polymerase polymers increased with age in the brains of Alzheimer mouse model and β-amyloid-activated PARP polymers inducing astrocytic metabolic failure and neuronal death in response to oxidative stress.[63] Poly(ADP-ribosyl) polymerase polymorphism is shown to modify the risk of AD in both an independent manner and interaction with proinflammatory interleukin-1A gene.[64] Poly(ADP-ribosyl) polymerase gene is highly associated with AD susceptibility. The distributions of PARP haplotype, Ht3-TT, and Ht4-CC were significantly associated with an increased risk of AD, whereas the Ht1-TC haplotype showed a protective effect for AD when compared with the control participants.[65] Conversely, thiamine has a cytoprotective effect on cultured neonatal rat cardiomyocytes under hypoxic insult by inhibiting PARP cleavage and DNA fragmentation.[66] Furthermore, benfotiamine prevents liposaccharide-induced apoptosis and PARP cleavage in mouse macrophage cell lines.[67] The relationship between thiamine and AD has been summarized in Table 1.

Histopathogenesis

The pathogenesis of AD is complex. Several factors, including amyloid plaques, neurofibrillary tangles, the loss of neuronal cells, and inflammatory processes are likely to contribute to the development of the disease.

Amyloid Plaques and Neurofibrillary Tangles

The accumulation of β-amyloid protein is considered to be a main cause of neuronal cell death in AD. Amyloid precursor protein protects against excito-toxicity following neuronal injuries by regulating the function of the glial glutamate transporters. Prolonged treatment with APP has been shown to augment glutamate clearance by cultured astrocytes and induce a dramatic decrease in glutamatergic synaptic activity of neurons cocultured with astrocytes.[68] In postmortem tissue, neurofibrillary tangles have been found in the nucleus basalis of chronic alcoholics.[69] Thiamine deficiency in mice led to the accumulation of APP within neuritic clusters in damaged brain regions.[70,71] Abnormal clusters were found to occur only in areas of neuronal damage, for example, the thalamus, mammillary bodies, and inferior colliculus. These clusters appeared as either irregular clumps or round or oval rosettes that resembled the neuritic component of the amyloid plaques seen in AD.[71]

Loss of Neuronal Cells

In patients with AD, β-amyloid activates macrophages and microglia, producing inflammation that accelerates neuronal damage.[72] On the other hand, thiamine-deficient encephalopathy is characterized by a selective loss of neurons in the midbrain, thalamus, and cerebellum.[73] In addition, thiamine deficiency also produced fiber cell degeneration in mouse lenses.[74]

Inflammatory Processes

It is widely accepted that microglial-mediated inflammation contributes to the progression of AD.[72] Similarly, microglial activation has been shown to occur in the early and late stages of thiamine-deficient animal models.[75-78]

Neurotransmitters

Acetylcholine (ACh) and norepinephrine (NorEpi) are the most common neurotransmitters associated with the pathophysiological conditions observed in AD. It is hypothesized that these neurotransmitters are hypoactive in AD.

Acetylcholine

Regions throughout the neocortex receive cholinergic inputs from the basal forebrain. Neurotransmission mediated by ACh has been shown to play an important role in attention processing.[79] Moreover, cortical deficiencies in cholinergic neurotransmission are known to contribute to the characteristic cognitive deficits associated with AD.[79] It has been shown that choline acetyltransferase activity, synaptic reuptake of choline, and ACh synthesis are reduced in AD.[80,81] Alterations in the density of cholinergic and noncholinergic receptors for glutamate, noradrenaline, and serotonin have been reported in transgenic Tg2576 mice with β-amyloid plaque pathology.[82] On the other hand, it has been suggested that thiamine-deficient encephalopathy involves impairment in cholinergic neurotransmitter function. Thiamine is an acoenzyme required for the synthesis of ACh, which is the neurotransmitter that is most commonly deficient in AD. In the neocortex, impaired coupling of muscarinic M_1 receptors to G-proteins has been shown to be associated with the severity of dementia in AD.[83] Moreover, it has been shown that the synthesis of ACh is impaired in the brains of thiamine-deficient rats,[84] leading to a significant reduction of neuronal ACh levels.[85] Animal studies have also suggested that thiamine is involved in the presynaptic release of ACh; it has been shown that thiamine binds to nicotinic receptors and may exhibit anticholinesterase activity.[86] In addition, thiamine deficiency has been shown to induce an early central muscarinic cholinergic lesion.[87]

Norepinephrine

The locus coeruleus (LC) is the major source of NorEpi, which is known to reduce neuroinflammation in the brain. Decreases in the number of NorEpi-containing neurons in the LC suggests that there should be reduced NorEpi activity in patients with AD.[88] Similarly, the concentration of NorEpi within CSF of participants with thiamine-deficient encephalopathy are also significantly reduced.[89] Using biochemical analyses, Mair et al[90] demonstrated that the concentration of NorEpi was significantly reduced in the brain (cortex, hippocampus, and olfactory bulb) accompanied by a concomitant decrease in behavioral measures of learning and memory in thiamine-deficient rats, suggesting that NorEpi deficits may contribute to the memory impairment. Moreover, Mousseau et al[91] demonstrated a brain region-selective vesicular dysfunction in catecholamine metabolism in an experimental thiamine deficiency model and suggested that it may account for certain neurobehavioral effects commonly encountered in thiamine deficiency. Furthermore, thiamine and its derivative (allylthiamindisulfide) significantly increased the plasma concentrations of noradrenaline and adrenaline in rats.[92]

Glutamate

Glutamate is synthesized from glucose and glutamine in presynaptic neuronal terminals and serves as a major excitatory neurotransmitter in the central nervous system (CNS). The capacity for cognition and memory is derived from various input and output pathways between the hippocampus and neocortex that rely on glutamatergic signaling.[93] Anomalies in

Table 1. The Relationship Between Thiamine and Alzheimer's Disease

	Alzheimer's Disease	Thiamine
Transketolase		
TK has been used to assess thiamine activity in mammalian tissue.	Structural abnormalities of TK occur in a high proportion of patients with AD.	TK variants have been found in patients with thiamine-deficient encephalopathy.
	Fibroblasts from patients with AD have been shown to exhibit TK abnormalities and might be used as a biochemical marker in patients with AD.	
Apolipoprotein E		
	ApoE is a major genetic factor identified in AD.	Frequency of the *ApoE* ε*4* allele is significantly higher in patients with thiamine-deficient dementia.
	Carriers of at least one *ApoE* ε*4* allele have an increased risk of developing AD.	
	ApoE has been shown to be significantly altered in the CSF of patients with AD.	
	Capillary cerebral amyloid angiopathy has been identified as a distinct ApoE ε4-associated subtype of sporadic AD.	
	Patients expressing ApoE genotype are known to have significantly impairment in memory retention.	
α-1-Antitrypsin		
	ATT has been identified as a potential biomarker in the plasma of patients with AD.	ATT polymorphism is associated with increased incidence of thiamine deficiency.
	High CSF ATT levels were associated with the clinical diagnosis of AD.	
Pyruvated dehydrogenase		
	Genetic defects in PDH complex are known to cause lactic acidosis, neurological deficits, and premature death. This defect show reduced PDH E1-alpha subunit and decreased affinity of PDH complex for TDP.	Thiamine treatment is very effective in some patients with PDHC-deficient.
		Thiamine deficiency has been shown to reduce the mRNA levels of TK and PDH.
p53 and glycogen synthetase kinase-3β		
p53 involved in cell cycle arrest or apoptosis.	In cellular models of AD, p53 has been found to be undergoing conformational changes that make cells less vulnerable to stressors or genocitoxic insults.	TDP has been shown to inhibit p53 binding.
GSK3β is a protein kinase involved in many physiological processes, for example, cell structure, metabolism, gene expression, and apoptosis	DNA damage has been shown to increase p53 DNA binding activity.	Thiamine has been shown to inhibit intracellular p53 activity.
Interaction of p53 and GSK3β has been shown to lead to increased activity of both proteins.	p53 and GSK3β have been linked to AD.	In an animal AD model, benfotiamine was found to improve the cognitive function and suppress GSK3 activity.
c-Fos		
c-Fos is involved in growth-related transcriptional control	Activation of c-Fos has been suggested to contribute to β-amyloid protein-induced neurotoxicity in AD.	c-Fos expression has been shown to increase with the progression of thiamine deficiency and associated with neuronal loss in the thalamus and inferior colliculus of the thiamine-deficient rat.
	Overexpression of c-Fos mRNA has been reported in AD.	

(continued)

Table 1. (continued)

	Alzheimer's Disease	Thiamine
Sp1		
	Sp1 transcription factor may be involved in regulating the expression of several AD-related proteins. Abnormal Sp1 transcription factor has been reported in AD.	The expression of the genes encoding thiamine transporters (THTR-1 and THTR02) are regulated via Sp1.
PARP		
PARP contributes to both neuronal death and survival under stress condition.	PARP cleavage is enhanced in peripheral blood mononuclear cells from patients with mild cognitive impairment. Enhanced PARP activity is reported in AD and is suggested as a marker for AD. PARP gene is highly associated with AD susceptibility.	Thiamine inhibits PARP cleavage and DNA fragmentation. Benfotiamine also prevents PARP cleavage in mouse macrophage cell line.

Abbreviations: PARP, poly(ADP-ribosyl) polymerase; AD, Alzheimer's disease; THTR, thiamine transporter; mRNA, messenger RNA; GSK3β, glycogen synthetase kinase-3β; TDP, thiamine diphosphate; TK, transketolase; PDH, pyruvated dehydrogenase; ATT, alpha-1-antitrypsin; ApoE, apolipoprotein E.

glutamate homeostasis may contribute to the pathological processes involved in AD. Uptake by the vesicular glutamate transporter has been shown to be decreased in patients with AD.[94] Glutamate transporters are believed to protect neurons against excito-toxicity by removing extracellular glutamate, and the β-amyloid protein prevents excito-toxicity via the recruitment of glial glutamate transporters.[95] In transgenic models of early-stage amyloid pathology, a significant reduction in the density of cholinergic, glutamatergic, and GABAergic presynaptic boutons has been observed.[96] In addition, alterations in the expression of glutamatergic transporters and receptors have been reported in cases of sporadic AD.[97] Studies have shown that decreased function of glutamate transporters in AD might lead to neurodegeneration. The excitatory amino acid transporter-2 (EAAT-2), a glutamate transporter, is expressed in astrocytes. In AD, many EAAT-2-positive neurons have been shown to have abnormalities in the cytoskeleton and the tau-protein. Moreover, these neurons were often found to have condensed and shrunken nuclei.[98] Neuronal glutamate transporter and EAAT-2 mRNA splice variants have been found in the brain of patients with AD.[99,100] In addition, a significant reduction in EAAT2 protein expression levels has been reported in the mid-frontal cortex of patients with AD.[101] Glial EAAT-1 was shown to be selectively expressed in degenerating neurons and dystrophic neuritis in AD.[102] Altered glutamate transport and aberrant EAAT-1 expression were found to be decreased in patients with AD brains.[103,104] On the other hand, glutamate transporters have been shown to be downregulated in thiamine-deficient astrocytes.[105,106] It has also been shown that the levels of EAAT-1 and EAAT-2 are diminished by 62% and 71%, respectively, in individuals with thiamine-deficient encephalopathy and treatment with *N*-acetylcysteine also prevented the downregulation of EAAT-2 in the medial thalamus and ameliorated the loss of several other astrocytic proteins.[107] In addition, treatment with pyrithiamine, a central thiamine antagonist, has been shown to decrease the protein levels of astrocytic glutamate transporters in the medial thalamus.[108]

Histamine

Histamine (HA) has a wide spectrum of biological actions in the CNS. Increased levels of HA have been noted in the brain, serum, and CSF of patients with AD.[109,110] Other studies, however, have shown a decrease in HA content in the hypothalamus, hippocampus, and temporal cortex of AD brains.[111,112] In severe cases of AD, an increased density of H3 HA-receptors has been detected in the medial temporal cortex.[112] The novel H3 HA-receptor antagonist (GSK189254) has been shown to increase the release of ACh, noradrenaline, and dopamine in the anterior cingulate cortex and ACh in the dorsal hippocampus. In addition, GSK189354 was found to significantly improve the performance of rats in a number of cognition testing paradigms.[113] Similarly, increased HA release was observed in the lateral thalamus in thiamine-deficient rats.[114] These rats demonstrated mouse-killing aggression (muricide). In addition, H1 HA-receptor antagonists (diphenhydramine, promethazine, and chlorpheniramine) have been shown to suppress muricide.[115] In thiamine-deficient rats, HA levels were found to be lower in the hippocampus, amygdala, olfactory bulb, thalamus, and pons-medulla; but, higher HA levels were observed in the hypothalamus.[116] After dietary thiamine supplements, however, these HA levels returned to normal in these rats. The relationship between thiamine and AD on the neurotransmitter has been summarized in Table 2.

Role of Thiamine in AD

Thiamine Trials

In patients with AD, fursultiamine, a derivative of thiamine, was found to have a mild beneficial effect using an oral dose of 100 mg/d in a 12-week trial.[117] In a short-term trial using 3 g/d of oral thiamine, Blass et al[118] demonstrated that the global cognitive rating by the Mini-Mental State Examination was improved in patients with AD. In another study using 3 to 8 g/d of oral thiamine, Meador et al[119] reported a mild beneficial

Table 2. Thiamine and Neurotransmitters

	Alzheimer's Disease	Thiamine
Acetylcholine		
Acetylcholine has been shown to play an important role in attention processing.	Cortical deficiencies in cholinergic neurotransmission are known to contribute to the characteristic cognitive deficits associated with AD.	Thiamine-deficient encephalopathy involves impairment in cholinergic neurotransmitter function.
	Choline acetyltransferase activity and ACh synthesis are reduced in AD.	Thiamine is a coenzyme required for the synthesis of ACh.
	In the neocortex, impaired coupling of muscarinic M_1 receptors to G-proteins has been shown to be associated with the severity of dementia in AD.	The synthesis of ACh is impaired in the brains of thiamine-deficient rats.
		Thiamine deficiency has been shown to induce an early central muscarinic cholinergic lesion.
Norepinephrine		
	Reduced NorEpi activity in patients with AD.	The concentration of NorEpi was significantly reduced in the brain (cortex, hippocampus and olfactory bulb) accompanied by a concomitant decrease in behavioral measures of learning and memory in thiamine-deficient rats
Glutamate		
	Uptake by the vesicular glutamate transport has been shown to be decreased in patients with AD.	Glutamate transporters have been shown to be downregulated in thiamine-deficient astrocytes.
	In AD, many excitatory amino acid transporter-(EAAT-2, a glutamate transporter) positive neurons have been shown to have abnormalities in the skeleton and the tau-protein.	The levels of EAAT-1 and EAAT-2 are diminished by 62 and 71%, respectively, in individuals with thiamine-deficient encephalopathy.
	Neuronal glutamate transporter and EAAT-2 splice variants have been found in the brain of patients with AD.	Treatment with pyrithiamine, a central thiamine antagonist, has been shown to decrease the protein levels of astrocytic glutamate transporters in the medial thalamus.
	Altered glutamate transport and aberrant EAAT-1 expression were found to be decreased in patients with AD brains.	
Histamine		
	Increased levels of HA have been noted in the brain, serum and CSF of patients with AD.	In thiamine-deficient rats, HA levels were found to be lower in the hippocampus, amygdala, olfactory bulb, thalamus, and pons-medulla; but higher levels were observed in the hypothalamus. After dietary thiamine supplement, however, these HA levels returned to normal in these rats
	Other studies demonstrated a decrease in HA content in the hypothalamus, hippocampus, and temporal cortex of AD brains.	

Abbreviations: HA, histamine; AD, Alzheimer's disease; CSF, cerebrospinal fluid; EAAT-2, excitatory amino acid transporter-2; ACh, acetylcholine; NorEpi, norepinephrine.

effect of thiamine in patients with AD. The long-term oral administration of thiamine at 3 g/d, however, did not slow the progression of AD.[120] Eight weeks benfotiamine treatment was found to improve the cognitive function and reduce both number of amyloid plaques and phosphorylated tau levels via a thiamine-independent mechanism in an animal model of AD.[121] These effects, however, were not found using fursultiamine. Interestingly, benfotiamine was unable to raise the levels of intracerebral thiamine phosphate derivatives.[122] In addition, it was found that benfotiamine does not easily cross neuroblastoma cell membrane in cultured cells.[123] Sulbutiamine, a lipid-soluble thiamine disulfide derivative, that increases thiamine derivatives in the brain and in cultured cells, can be used as a CNS drug. In rats, chronic treatment with sulbutiamine was shown to improve memory in an object recognition task and reduce the amnesic effects of dizocipine, which blocks the *N*-methyl-D-aspartate glutamatereceptors.[123,124] Furthermore, sulbutiamine has been shown to act synergistically with acetylcholinesterase inhibitors in patients with early or moderate stage AD.[125]

Pharmacokinetics

The intestinal absorption of thiamine is sufficient in young people, but may be reduced with age.[126] Schaller and Holler[127] reported that intestinal alkaline phosphatase (ALP) is involved in the active thiamine absorption in the intestinal tract. Furthermore, Rindi et al[128] found that intestinal ALP can transphosphorylate thiamine to thiamine monophosphate during intestinal transport in rats. Without ALP, it was found that thiamine could not be transported into the lumen of the gastrointestinal tract.[129] In addition, it has been shown that the intestinal ALP activity declines significantly with age in mice.[130] Enzymatic activity of ALP in the duodenum was also found significantly higher in 5-month-old rats compared with other age groups, especially 2.5 weeks and 23 months old rats.[131] The decrease in the intestinal ALP activity in old rats has been attributed to the reduction in the number of enterocytes caused by the age-induced atrophy of the intestinal mucosa.[132] In addition, oral absorption of thiamine was found to be poor in older individuals.[133] In humans, single oral doses of thiamine higher than 2.5 to 5 mg are largely unabsorbed.[134,135] Nichols and Basu[136] reported that almost half of the elderly individuals (older than or equal to 65 years) had an effect of TPP on the TK activity above 14%, which is suggestive of thiamine-deficient state. It should be noted, however, that the daily thiamine intake of these patients was above the recommended requirement (>0.4 mg/1000 kcal). Baker et al[137] demonstrated that thiamine deficiency in individuals older than 60 years could only be corrected by intramuscular administration of thiamine. Sasaki et al[138] reported a case of thiamine deficiency with psychotic symptoms. The patient's condition was ameliorated only by repeated administrations of intravenous thiamine.

Conclusion

The relationship between thiamine and AD has been discussed, but it could not rule out the mass effect changes from cerebral atrophy in AD. Thiamine levels and activity of thiamine-dependent enzymes are reduced in the brains and peripheral tissues of patients with AD. Genetic studies, however, have provided the opportunity to determine what proteins link thiamine to AD pathology. Although oral thiamine supplement have been shown to improve cognitive function in patients with AD; but, thiamine absorption decreases with advancing age. In the early stages of thiamine-deficient encephalopathy (Wernicke's encephalopathy), patients were found to respond rapidly to large doses of parental thiamine. The initial dose of thiamine is usually at 100 mg 2 to 3 times daily for 1 to 2 weeks. Therefore, further studies to determine the therapeutic value of parental thiamine for treating AD would be of great interest.

Declaration of Conflicting Interests

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References

1. Siebert G, Gessner B, Klasser M. Energy supply of the central nervous system. In: Somogyi JC, Hotzel D, eds. *Nutrition and Neurobiology*. Basel: Karger; 1986:1-26.
2. Martin P, Singleton CK, Hiller-Stumhöfel S. The role of thiamine deficiency in alcoholic brain disease. *Alcohol Res Health*. 2003; 27(2):174-181.
3. Ishii K, Sarai K, Sanemori H, Kawasaki T. Concentration of thiamine and its phosphate esters in rat tissues determined by high pressure liquid chromography. *J Nutr Sci Vitaminol (Tokyo)*. 1979;25(6):517-523.
4. Bubber P, Ke ZJ, Gibson GE. Tricarboxylic acid cycle enzymes following thiamine deficiency. *Neurochem Int*. 2004;45(7): 1021-1028.
5. Mastrogiacomo F, Bergeron C, Kish SJ. Brain α-ketoglutarate dehydrogenases complexes activity in Alzheimer's disease. *J Neurochem*. 1993;61(6):2007-2014.
6. Ott A, Stolk RP, van Harskamp F, Pols HA, Hofman A, Breteler MM. Diabetes mellitus and the risk of dementia: the Rotterdam study. *Neurology*. 1999;53(9):1937-1942.
7. Kroner Z. The relationship between Alzheimer's disease and diabetes: type 3 diabetes? *Altern Med Rev*. 2009;14(4):373-379.
8. Takeda S, Sato N, Uchio-Yamada K, et al. Diabetes-accelerated memory dysfunction via cerebrovascular inflammation and abeta deposition in an Alzheimer mouse model with diabetes. *Proc Natl Acad Sci USA*. 2010;107(15):7036-7041.
9. Saito N, Kimura M, Kuchiba A, Itokawa Y. Blood thiamine levels in outpatients with diabetes mellitus. *J Nutr Sci Vitaminol*. 1987; 33(6):421-430.
10. Kjøsen BMS, Seim SH. The transketolase assay of thiamine in some diseases. *Am J Clin Nutr*. 1977;30(10):1591-1596.
11. Thornalley PJ, Babaei-Jadidi R, Al Ali H, et al. High prevalence of low thiamine concentration in diabetes linked to a marker of vascular disease. *Diabetologia*. 2007;50(10):2164-2170.
12. Glasø M, Nordbø G, Diep L, Bøhmer T. Reduced concentrations of several vitamins in normal weight patients with late-onset dementia of the Alzheimer type without vascular disease. *J Nutr Health Aging*. 2004;8(5):407-413.
13. Gold M, Chen MF, Johnson K. Plasma and red blood cell thiamine deficiency in patients with dementia of the Alzheimer's type. *Arch Neurol*. 1995;52(11):1081-1086.
14. Molina JA, Jiménez-Jiménez FJ, Hernánz A, et al. Cerebrospinal fluid levels of thiamine in patients with Alzheimer's disease. *J Neural Transm*. 2002;109(7-8):1035-1044.
15. Gibson GE, Sheu KF, Blass JP, et al. Reduced activities of thiamine-dependent enzymesin the brains and peripheral tissues of patients with Alzheimer's disease. *Arch Neurol*. 1988;45(8): 836-840.
16. Sheu KF, Clarke DD, Kim YT, Harding BJ, DeCicco J. Studies of transketolase abnormality in Alzheimer's disease. *Arch Neurol*. 1988;45(8):841-845.
17. Paoletti F, Mocali A, Marchi M, Sorbi S, Piacentini S. Occurrence of transketolase abnormalities in extracts of forskin fibroblasts from patients with Alzheimer's disease. *Biochem Biophys Res Commun*. 1990;172(2):396-401.

18. Tombaccini D, Mocali A, Paoletti F. Characteristic transketolase alterations in dermal fibroblasts of Alzheimer patients are modulated by culture conditions. *Exp Mol Pathol*. 1994;60(2):140-146.
19. Paoletti F, Mocali A, Tombaccini D. Cysteine proteinases are responsible for characteristic transketolase alterations in Alzheimer fibroblasts. *J Cell Physiol*. 1997;172(1):63-68.
20. Martin PR, McCool BA, Singleton CK. Molecular genetics of transketolase in the pathogenesis of the Wernicke-Korsakoff syndrome. *Metab Brain Dis*. 1995;10(1):45-55.
21. Blass JP, Gibson GE. Abnormality of a thiamine-requiring enzyme in patients with Wernicke-Korsakoff syndrome. *N Engl J Med*. 1977;297(25):1367-1370.
22. Mukherjee AB, Svoronos S, Ghazanfari A, et al. Transketolase abnormality in cultured fibroblasts from familial chronic alcoholic men and their male offspring. *J Clin Invest*. 1987;79(4):1039-1043.
23. Heap LC, Pratt OE, Ward RJ, et al. Individual susceptibility to Wernicke-Korsakoff syndrome and alcoholism-induced cognitive deficit: impaired thiamine utilization found in alcoholics and alcohol abusers. *Psychiatr Genet*. 2002;12(4):217-224.
24. Licastro F, Chiappelli M, Thal LJ, Masliah E. α-1-antichymotrypsin polymorphism in the gene promoter region affects survival and synapsis loss in Alzheimer's disease. *Arch Gerontol Geriatr Suppl*. 2004;(9):243-251.
25. Lane RM, Farlow MR. Lipid homeostatis and apolipoprotein E in the development and progression of Alzheimer's disease. *J Lipid Res*. 2005;46(5):949-968.
26. Puchades M, Hansson SF, Nilsson CL, Andreasen N, Blennow K, Davidsson P. Proteomic studies of potential cerebrospinal fluid protein markers for Alzheimer's disease. *Brain Res Mol Brain Res*. 2003;118(1-2):140-146.
27. Thal DR, Papassotiropoulos A, Saido TC, et al. Capillary cerebral amyloid angiopathy identifies a distinct APOE ε4-associated subtype of sporadic Alzheimer's disease. *Acta Neuropathol*. 2010;120(2):169-183.
28. Wolk DA, Dickerson BC. Alzheimer's disease neuroimaging initiative. Apolipoprotein E (APOE) genotype has dissociable effects on memory and attentional-executive network function in Alzheimer's disease. *PNAS*. 2010;107(22):10256-10261.
29. Paz-y-Miño C, Carrera C, López-Cortés A, et al. Genetic polymorphisms in apolipoprotein E and glutathione peroxidase 1 genes in the Ecuadorian population affected with Alzheimer's disease. *Am J Med Sci*. 2010;340(5):373-377.
30. Muramatsu T, Kato M, Matsui T, et al. Apolipoprotein E ε4 allele distribution in Wernicke-Korsakoff syndrome with or without global intellectual deficits. *J Neural Transm*. 1997;104(8-9):913-920.
31. Song F, Poljak A, Smythe GA, Sachdev P. Plasma biomarkers for mild cognitive impairment and Alzheimer's disease. *Brain Res Rev*. 2009;61(2):69-80.
32. Nielsen HM, Minthon L, Londos E, et al. Plasma and CSF serpins in Alzheimer disease and dementia with Lewy bodies. *Neurology*. 2007;69(16):1569-1579.
33. Schmechel DE, Browndyke J, Ghio A. Strategies for dissecting genetic-environmental interactions in neurodegenerative disorders. *Neurotoxicology*. 2006;27(5):637-657.
34. Schmechel DE. Art, α-1-antitrypsin polymorphisms and intense creative energy: blessing or curse? *Neurotoxicology*. 2007;28(5):899-914.
35. Jacobia SJ, Korotchkina LG, Patel MS. Characterization of a missense mutation at histidine-44 in a pyruvate dehydrogenase-deficient patient. *Biochim Biophys Acta*. 2002;1586(1):32-42.
36. Naito E, Ito M, Takeda E, Yokota I, Yoshijima S, Kuroda Y. Molecular analysis of abnormal pyruvate dehydrogenase in a patient with thiamine-responsive congenital lactic academia. *Pediatr Res*. 1994;36(3):340-346.
37. Naito E, Ito M, Yokota I, et al. Thiamine-responsive pyruvate dehydrogenase deficiency in two patients caused by a point mutation (F205L and L216F) within the thiamine pyrophosphate pyrophosphate binding region. *Biochim Biophys Acta*. 2002;1588(1):79-84.
38. Bonne G, Benelli C, De Meirleir L, et al. E1 pyruvate dehydrogenase deficiency in a child with motor neuropathy. *Pediatr Res*. 1993;33(3):284-288.
39. Meléndez RR. Importance of water-soluble vitamins as regulatory factors of genetic expression [in Spanish]. *Rev Invest Clin*. 2002;54(1):77-83.
40. Ko LJ, Prives C. p53: puzzle and paradigm. *Genes Dev*. 1996;10(9):1054-1072.
41. Lanni C, Racchi M, Mazzini G, et al. Conformationally altered p53: a novel Alzheimer's disease marker? *Mol Psychiatry*. 2008;13(6):641-647.
42. Uberti D, Lanni C, Carsana T, et al. Identification of a mutant-like conformation of p53 in fibroblasts from sporadic Alzheimer's disease patients. *Neurobiol Aging*. 2006;27(9):1193-1201.
43. McLure KG, Takagi M, Kastan MB. NAD^+ modulates p53 DNA binding specificity and function. *Mol Cellular Biol*. 2004;24(22):9958-9967.
44. Proctor CJ, Gray DA. GSK3 and p53—is there a link in Alzheimer's disease? *Mol Neurodegeneration*. 2010;5:7.
45. Hooper C, Killick R, Lovestone S. The GSK3 hypothesis of Alzheimer's disease. *J Neurochem*. 2008;104(6):1433-1439.
46. Zhao J, Sun X, Yu Z, et al. Exposure to pyrithiamine increases beta-amyloid accumulation, Tau hyperphosphorylation, and glycogen synthetase kinase-3 activity in the brain. *Neurotox Res*. 2010;19(4):575-583.
47. Pan X, Gong N, Zhao J, et al. Powerful beneficial effects of benfotiamine on cognitive impairment and beta-amyloid deposition in amyloid precursor protein/presenilin-1 transgenic mice. *Brain*. 2010;133(pt 5):1342-1351.
48. Gillardon F, Skutella T, Uhlmann E, Holsboer F, Zimmermann M, Behl C. Activation of c-Fos contributes to amyloid beta-peptide-induced neurotoxicity. *Brain Res*. 1996;706(1):169-172.
49. Lu W, Mi R, Tang H, Liu S, Fan M, Wang L. Over-expression of c-fos mRNA in the hippocampal neurons in Alzheimer's disease. *Chin Med J (Engl)*. 1998;111(1):35-37.
50. Marcus DL, Strafaci JA, Miller DC, et al. Quantitative neuronal c-fos and c-jun expression in Alzheimer's disease. *Neurobiol Aging*. 1998;19(5):393-400.
51. Hazell AS, Todd KG, Butterworth RF. Mechanisms of neuronal cell death in Wernicke's encephalopathy. *Metab Brain Dis*. 1998;13(2):97-122.

52. Hazell AS, McGahan L, Tetzlaff WW, et al. Immediate-early gene expression in the brain of the thiamine-deficient rat. *J Mol Neurosci*. 1998;10(1):1-15.
53. Zimitat C, Nixon PF. Glucose induced IEG expression in the thiamin-deficient rat brain. *Brain Res*. 2001;892(1):218-227.
54. Nabokina SM, Said HM. Characterization of the 5'-regulatory region of the human thiamin transporter SLC19A3: in vitro and in vivo studies. *Am J Physiol Gastrointest Liver Physiol*. 2004; 287(4):G822-G829.
55. Santpere G, Nieto M, Puig B, Ferrer I. Abnormal Sp1 transcription factor expression in Alzheimer's disease and tauopathies. *Neurosci Lett*. 2006;397(1-2):30-34.
56. Citron BA, Dennis JS, Zeitlin RS, Echeverrie V. Transcription factor Sp1 dysregulation in Alzheimer's disease. *J Neurosci Res*. 2008;86(11):2499-2504.
57. Brock B, Basha R, DiPalma K, et al. Co-localization and distribution of cerebral APP and Sp1 and its relationship to amyloidogenesis. *J Alzheimers Dis*. 2008;13(1):71-80.
58. Christensen MA, Zhou W, Qing H, Lehman A, Philipsen S, Song W. Transcriptional regulation of BACE1, the β-amyloid precursor protein β-secretase, by Sp1. *Mol Cell Biol*. 2004;24(2):865-874.
59. Docagne F, Gabriel C, Lebeurrier N, et al. Sp1 and Smad transcription factors co-operate to mediate TGF-β-dependent activation of amyloid-β precursor protein gene transcription. *Biochem J*. 2004; 383(pt 2):393-399.
60. Gatta L, Cardinale A, Wannenes F, et al. Peripheral blood mononuclear cells from mild cognitive impairment patients show deregulation of Bax and Sods1 mRNAs. *Neurosci Lett*. 2009;453(1):36-40.
61. Love S, Barber R, Wilcock GK. Increased poly(ADP-ribosyl)ation of nuclear proteins in Alzheimer's disease. *Brain*. 1999; 122(pt 2):247-253.
62. Kassner SS, Bonaterra GA, Kaiser E, et al. Novel systemic markers for patients with Alzheimer disease?–a pilot study. *Curr Alzheimer Res*. 2008;5(4):358-366.
63. Abeti R, Abramov AY, Duchen MR. β-Amyloid activates PARP causing astrocytic metabolic failure and neuronal death. *Brain*. 2011;134(pt 6):1658-1672.
64. Infante J, Llorca J, Mateo I, et al. Interaction between poly(ADP-ribose) polymerase 1 and interleukin 1A genes is associated with Alzheimer's disease risk. *Dement Geriatr Cogn Disord*. 2007; 23(4):215-218.
65. Liu HP, Lin WY, Wu BT, et al. Evaluation of the poly(ADP-ribose) polymerase-1 gene variants in Alzheimer's disease. *J Clin Lab Anal*. 2010;24(3):182-186.
66. Shin BH, Choi SH, Cho EY, et al. Thiamine attenuates hypoxia-induced cell death in cultured neonatal rat cardiomyocytes. *Mol Cells*. 2004;18(2):133-140.
67. Yadav UCS, Kalariya NM, Srivastava SK, Ramana KV. Protective role of benfotiamine, a fat soluble vitamin B1 analogue, in the liposaccharide-induced cytotoxic signals in murine macrophages. *Free Radic Biol Med*. 2010;48(10):1423-1434.
68. Ikegaya Y, Matsura S, Ueno S, et al. β-amyloid enhances glial glutamate uptake activity and attenuates synaptic efficacy. *J Biol Chem*. 2002;277(35):32180-32186.
69. Cullen KM, Hallen GM. Neurofibrillary tangles in chronic alcoholics. *Neuropathol Appl Neurobiol*. 1995;21(4):312-318.
70. Calingasan NY, Gandy SE, Baker H, et al. Accumulation of amyloid precursor protein-like immunoreactivity in rat brain in response to thiamine deficiency. *Brain Res*. 1995;677(1): 50-60.
71. Calingasan NY, Gandy SE, Baker H, et al. Novel neuritic clusters with accumulations of amyloid precursor protein and amyloid precursor-like protein 2 immunoreactivity in brain regions damaged by thiamine deficiency. *Am J Pathol*. 1996;149(3): 1063-1071.
72. Mandrekar-Colucci S, Landreth GE. Microglia and inflammation in Alzheimer's disease. *CNS Neurol Disord Drug Targets*. 2010; 9(2):156-167.
73. Karuppagounder SS, Xu H, Shi Q, et al. Thiamine deficiency induces oxidative stress and exacerbates the plaque pathology in Alzheimer's mouse model. *Neurobiol Aging*. 2009;30(10): 1587-1600.
74. Frederikse PH, Farnsworth P, Zigler JS. Thiamine deficiency in vivo produces fiber cell degeneration in mouse lenses. *Biochem Biophys Res Commun*. 1999;258(3):703-707.
75. Hazell AS, Sheedy D, Oanea R, Aghourian M, Sun S, Jung JY, et al. Loss of astrocytic glutamate transporters in Wernicke encephalopathy. *Glia*. 2010;58:148-156.
76. Ke ZJ, Gibson GE. Selective response of various brain cell types during neurodegeneration induced by mild impairment of oxidative metabolism. *Neurochem Int*. 2004;45(2-3):361-369.
77. Todd KG, Butterworth RF. Early microglial response in experimental thiamine deficiency: an immunohistochemical analysis. *Glia*. 1999;25(2):190-198.
78. Calingasan NY, Park LC, Calo LL, Trifiletti RR, Gandy SE, Gibson GE. Induction of nitric oxide synthase and microglial responses precede selective cell death induced by chronic impairment of oxidative metabolism. *Am J Pathol*. 1998;153(2): 599-610.
79. Francis PT, Palmer AM, Snape M, Wilcox GK. The cholinergic hypothesis of Alzheimer's disease: a review of progress. *J Neurol Neurosurg Psychiatry*. 1999;66(2):137-147.
80. Sims NR, Bowen DM, Allen SJ, et al. Presynaptic cholinergic dysfunction in patients with dementia. *J Neurochem*. 1983; 40(2):503-509.
81. Francis PT, Palmer AM, Sims NR, et al. Neurochemical studies of early-onset Alzheimer's disease. Possible influence on treatment. *N Engl J Med*. 1985;313(1):7-11.
82. Klingner M, Apelt J, Kumar A, et al. Alterations in cholinergic and non-cholinergic neurotransmitter receptor densities in transgenic Tg2576 mouse brain with beta-amyloid plaque pathology. *Int J Dev Neurosci*. 2003;21(7):357-369.
83. Tsang SW, Lai MK, Kirvell S, et al. Impaired coupling of muscarinic M1 receptors to G-proteins in the neocortex is associated with severity of dementia in Alzheimer's disease. *Neurobiol Aging*. 2006;27(9):1216-1223.
84. Ruenwongsa P, Pattanavibag S. Impairment of acetylcholine synthesis in deficient rats developed by prolonged tea consumption. *Life Sci*. 1984;34(4):365-370.
85. Vorhees CV, Schmidt DE, Barrett RJ. Effects of pyrithiamin on acetylcholine levels and utilization in rat brain. *Brain Res Bull*. 1978;3(5):493-496.

86. Meador KJ, Nichols ME, Franke P, et al. Evidence for a central cholinergic effect of high-dose thiamine. *Ann Neurol*. 1993; 34(5):724-726.
87. Barclay LL, Gibson GE, Blass JP. Impairment of behavior and acetylcholine metabolism in thiamine deficiency. *J Pharmacol Exp Ther*. 1981;217(3):537-543.
88. Heneka MT, Nadrigny F, Regen T, et al. Locus ceruleus controls Alzheimer's disease pathology by modulating microglial functions through norepinephrine. *Proc Natl Acad Sci USA*. 2010; 107(13):6058-6063.
89. McEntee WJ, Mair RG. Memory impairment in Korsakoff's psychosis: a correlation with brain noradrenergic activity. *Science*. 1978;202(4370):905-907.
90. Mair RG, Anderson CD, Langlais PJ, McEntee WJ. Thiamine deficiency depletes cortical norepinephrine and impairs learning processes in the rat. *Brain Res*. 1985;360(1-2): 273-284.
91. Mousseau DD, Raghavendra Rao VL, Butterworth RF. Vesicular dysfunction during experimental thiamine deficiency is indicated by alterations in dopamine metabolism. *Eur J Pharmacol*. 1996;317(2-3):263-267.
92. Oi Y, Shishido C, Wada K, Odaka H, Ikeda H, Iwai K. Allylthiamindisulfide and related compounds enhance thermogenesis with increasing noradrenaline and adrenaline secretion in rats. *J Nutr Sci Vitaminol*. 1999;45(5):643-653.
93. Squire LR, Zola-Morgan S. The medial temporal lobe memory system. *Science*. 1991;253(5026):1380-1386.
94. Westphalen RI, Scott HL, Dodd PR. Synaptic vesicle transport and synaptic membrane transporter sites in excitatory amino acid nerve terminals in Alzheimer Disease. *J Neural Transm*. 2003; 110(9):1013-1027.
95. Baba A, Mitsumori K, Yamada MK, Nishiyama N, Matsuki N, Ikegaya Y. Beta-amyloid prevents excitotoxicity via recruitment of glial glutamate transporters. *Naunyn Schmiedebergs Arch Pharmacol*. 2003;368(3):234-238.
96. Bell KF, Ducatenzeiler A, Ribeiro-da-Silva A, Duff K, Bennett DA, Cuello AC. The amyloid pathology progresses in a neurotransmitter-specific manner. *Neurobiol Aging*. 2006; 27(11):1644-1657.
97. Jacob CP, Koutsilieri E, Bartl J, et al. Alterations in expression of glutamatergic transporters and receptors in sporadic Alzheimer's disease. *J Alzheimers Dis*. 2007;11(1):97-116.
98. Thal DR. Excitatory amino acid transporter EAAT-2 in tangle-bearing neurons in Alzheimer's disease. *Brain Pathol*. 2002; 12(4):405-411.
99. Scott HL, Tannenberg AE, Dodd PR. Variant forms of neuronal glutamate transporter sites in Alzheimer's disease cerebral cortex. *J Neurochem*. 1995;64(5):2193-2202.
100. Honig LS, Chambliss DD, Bigio EH, Carroll SL, Elliott JL. Glutamate transporter EAAT2 splice variants occur not only in ASL, but also in AD and controls. *Neurology*. 2000;55(8): 1082-1088.
101. Li S, Mallory M, Alford M, Tanaka S, Masliah E. Glutamate transporter alterations in Alzheimer disease are possibly associated with abnormal APP expression. *J Neuropathol Exp Neurol*. 1997;56:901-911.
102. Scott HL, Pow DV, Tannenberg AE, Dodd PR. Aberrant expression of the glutamate transporter excitatory amino acid transporter 1 (EAAT1) in Alzheimer's disease. *J Neurosci*. 2002;22(3): RC206.
103. Masliah E, Alford M, Mallory M, Rockkenstein E, Moechars D, Van Leuven F. Abnormal transport function in mutant amyloid precursor protein transgenic mice. *Exp Neurol*. 2000;163(2): 381-387.
104. Zoia CP, Tagliabue E, Isella V, et al. Fibroblast glutamate transport in aging and in AD: correlations with disease severity. *Neurobiol Aging*. 2005;26(6):825-832.
105. Hazell AS, Todd KG, Butterworth RF. Mechanisms of neuronal cell death in Wernicke's encephalopathy. *Metab Brain Dis*. 1998;13(2):97-122.
106. Hazell AS, Pannunzio P, Rama Rao KV, Pow DV, Rambaldi A. Thiamine deficiency results in downregulation of the GLAST glutamate transporter in cultured astrocytes. *Glia*. 2003;43(2): 175-184.
107. Hazell AS, Sheedy D, Oanea R, et al. Loss of astrocytic glutamate transporters in Wernicke encephalopathy. *Glia*. 2010; 58(2):148-156.
108. Hazell AS, Rao KV, Danbolt NC, Pow DV, Butterworth RF. Selective down-regulation of the astrocyte glutamate transporters GLT-1 and GLAST within the medial thalamus in experimental Wernicke's encephalopathy. *J Neurochem*. 2001;78(3): 560-568.
109. Cacabelos R, Yamatodani A, Niigawa H, et al. Brain histamine in Alzheimer's disease. *Methods Find Exp Clin Pharmacol*. 1989;11(5):353-360.
110. Cacabelos R, Fernández-Novoa L, Pérez-Trullén JM, Franco-Maside A, Alvarez XA. Serum histamine in Alzheimer's disease and multi-infarct dementia. *Methods Find Exp Clin Pharmacol*. 1992;14(9):711-715.
111. Mazurkiewicz-Kwilecki IM, Nsonwah S. Changes in the regional brain histamine and histidine levels in postmortem brains of Alzheimer patients. *Can J Physiol Pharmacol*. 1989; 67(1):75-78.
112. Panula P, Rinne J, Kuokkanen K, et al. Neuronal histamine deficit in Alzheimer's disease. *Neuroscience*. 1998;82(4):993-997.
113. Medhurst AD, Atkins AR, Beresford IJ, et al. GSK189254, a novel H_3 receptor antagonist that binds to histamine H_3 receptors in Alzheimer's disease brain and improves cognitive performance in preclinical models. *J Pharmacol Exper Ther*. 2007; 321(3):1032-1045.
114. McRee RC, Terry-Ferguson M, Langlais PJ, et al. Increased histamine release and granulocytes within the thalamus of a rat model of Wernicke's encephalopathy. *Brain Res*. 2000;858(2): 227-236.
115. Onodera K, Ogura Y. Effects of histaminergic drugs on muricide induced by thiamine deficiency. *Jpn J Pharmacol*. 1984;34(1): 15-21.
116. Onodera K, Maeyama K, Watanabe T. Regional changes in brain histamine levels following dietary-induced thiamine deficiency in rats. *Jpn J Pharmacol*. 1990;54:339-343.
117. Mimori Y, Katsuoka H, Nakamura S. Thiamine therapy in Alzheimer's disease. *Metab Brain Dis*. 1996;11(1):89-94.

118. Blass JP, Gleason P, Brush D, DiPonte P, Thaler H. Thiamine and Alzheimer's disease. A pilot study. *Arch Neurol*. 1988; 45(8):833-835.
119. Meador K, Loring D, Nichols M, et al. Preliminary findings of high-dose thiamine in dementia of Alzheimer's type. *J Geriatr Psychiatry Neurol*. 1993;6(4):222-229.
120. Nolan KA, Black RS, Sheu KFR, Lanberg J, Blass JP. A trial of thiamine in Alzheimer's disease. *Arch Neurol*. 1991;48(1):81-83.
121. Pan X, Gong N, Zhao J, et al. Powerful beneficial effects of benfotiamine on cognitive impairment and beta-amyloid deposition in amyloidprecursor protein/presenilin-1 transgenic mice. *Brain*. 2010;133(pt 5):1342-1351.
122. Volvert ML, Seyen S, Piette M, et al. Benfotiamine, a synthetic S-acyl thiamine derivative, has different mechanisms of action and a different pharmacological profile than lipid-soluble thiamine disulfide derivatives. *BMC Pharmacol*. 2008;8:10.
123. Bizot JC, Herpin A, Pothion S, Pirot S, Trovero F, Ollat H. Chronic treatment with sulbutiamine improves memory in an object recognition task and reduces some amnesic effects of dizocilpine in a spatial delayed-non-match-to-sample task. *Prog Neuropsychopharmacol Biol Psychiatry*. 2005;29(6): 928-935.
124. Micheau J, Durkin TP, Destrade C, Rolland Y, Jaffard R. Chronic administration of sulbutiamine improves long term memory formation in mice: possible cholinergic mediation. *Pharmacol Biochem Behav*. 1985;23(2):195-198.
125. Ollat H, Laurent B, Bakchine S, Michel BF, Touchon J, Dubois B. Effects of the association of sulbutiamine with an acetylcholinesterase inhibitor in early stage and moderate Alzheimer disease [in French]. *Encephale*. 2007;33(2):211-215.
126. Baum RA, Iber FL. Thiamine–the interaction of aging, alcoholism, and malabsorption in various populations. *World Rev Nutr Diet*. 1984;44:85-116.
127. Schaller K, Holler H. Thiamine absorption in rat. II. Intestinal alkaline phosphatase activity and thiamine absorption from rat small intestines in-vitro and in-vivo. *Int J Vit Nutr Res*. 1975; 45(1):30-38.
128. Rindi G, Ricci V, Gastaldi G, Patrini C. Intestinal alkaline phosphatase can transphosphorylate during intestinal transport in the rat. *Arch Physiol Biochem*. 1995;103(1):33-38.
129. Luong KVQ, Nguyen LTH. Adult hypophosphatasia and a low level of red blood cell thiamine pyrophosphate. *Ann Nutr Metab*. 2005;49(2):107-109.
130. Detel D, Baticic L, Varljen J. The influence of age on intestinal dipeptidyl peptidase IV (DPP IV/CD26), disaccharidase, and alkaline phosphatase activities in C57BL/6 mice. *Exp Aging Res*. 2008;34(1):49-62.
131. Jang I, Jung K, Cho J. Influence of age on duodenal brush border membrane and specific activities of brush border membrane enzymes in Wistar rats. *Exp Anim*. 2000;49(4):281-287.
132. Höhn P, Gabbert H, Wagner R. Differentiation and aging of the rat intestinal mucosa. II. Morphological, enzyme histochemical and disc electrophoretic aspects of the aging of the small intestinal mucosa. *Mech Ageing Dev*. 1978;7(3):217-226.
133. Tallaksen CME, Sande A, Bøhmer T, Bell H, Karlsen J. Kinetic of thiamin and thiamin phosphate esters in human blood, plasma and urine after 50 mg intravenously or orally. *Eur J Clin Pharmacol*. 1993;44(1):73-78.
134. Friedman TE, Kmieckiak TC, Keegan PK, Sheft BB. The absorption, destruction, and excretion of orally administered thiamine by human subjects. *Gastroenterology*. 1948;11(1): 100-114.
135. Morrison AB, Campbell JA. Vitamin absorption studies. I. Factors influencing the excretion of oral test doses of thiamine and riboflavin by human subjects. *J Nutr*. 1960;72:435-440.
136. Nichols HK, Basu TK. Thiamine status of the elderly: dietary intake and thiamine pyrophosphate response. *J Am Coll Nutr*. 1994;13(1):57-61.
137. Baker H, Frank O, Jaslow SP. Oral versus intramuscular vitamin supplementation for hypovitaminosis in the elderly. *J Am Geriatr Soc*. 1980;28(1):42-45.
138. Sasaki T, Yukizane T, Atsuta H, et al. A case of thiamine deficiency with psychotic symptoms–blood concentration of thiamine and response to therapy [in Japanese]. *Seishin Shinkeigaku Zasshi*. 2010;112(2):97-110.